

Mastering Quantum Mechanics — Problem 9.9

Coherent states as displaced ground states

Solved by eattardo@gmail.com

Problem 9.9

Coherent states as displaced ground states. For $\psi_\lambda \equiv N e^{\lambda \hat{a}^\dagger} \varphi_0$ with $\lambda \in \mathbb{C}$ and φ_0 the normalized ground state: (1) find $N > 0$ for normalization; (2) compute $\hat{a}\psi_\lambda$ in terms of ψ_λ ; (3) find $\langle \hat{H} \rangle$; (4) find the energy uncertainty ΔE_λ ; (5) obtain and solve the ODE for $\psi_\lambda(u)$, $u = x/L_0$; (6) show $\psi_\lambda(u) \sim \varphi_0(u - b)$ and find b ; (7) interpret real vs. imaginary parts of λ .

Hints & Orientation

- **(1)** Taylor expand: $e^{\lambda \hat{a}^\dagger} \varphi_0 = \sum_n \frac{\lambda^n}{n!} \varphi_n$ using $(\hat{a}^\dagger)^n \varphi_0 = \sqrt{n!} \varphi_n$. Orthonormality turns $\langle \psi_\lambda, \psi_\lambda \rangle$ into $\sum_n |\lambda|^{2n}/n! = e^{|\lambda|^2}$.
- **(2)** Use $[\hat{a}, e^{\lambda \hat{a}^\dagger}] = \lambda e^{\lambda \hat{a}^\dagger}$ (from $[\hat{a}, \hat{a}^\dagger] = 1$, Eq. (9.3.10), or the lemma $[\hat{a}, (\hat{a}^\dagger)^k] = k(\hat{a}^\dagger)^{k-1}$) and $\hat{a}\varphi_0 = 0$. You should find ψ_λ is an *eigenstate* of $\hat{a}$.
- **(3)** Write $\hat{H} = \hbar\omega(\hat{a}^\dagger \hat{a} + \frac{1}{2})$ (Eq. (9.3.14)) and use $\hat{a}\psi_\lambda = \lambda\psi_\lambda$ together with its adjoint $\langle \psi_\lambda | \hat{a}^\dagger = \lambda^* \langle \psi_\lambda |$.
- **(4)** You need $\langle \hat{N}^2 \rangle$; normal-order $\hat{N}^2 = \hat{a}^\dagger \hat{a}^\dagger \hat{a} \hat{a} + \hat{a}^\dagger \hat{a}$ so every $\hat{a}$ acts to the right as λ .
- **(5)** In position space $\hat{a} = \frac{1}{\sqrt{2}}(u + \frac{d}{du})$; the eigenvalue equation becomes a first-order ODE.
- **(6)** Complete the square in the exponent and compare with $\varphi_0(u) \propto e^{-u^2/2}$ (Eq. (9.2.40)).
- **(7)** $\langle \hat{x} \rangle \propto \text{Re } \lambda$, $\langle \hat{p} \rangle \propto \text{Im } \lambda$.

Solution

Throughout, $\hat{a}\varphi_n = \sqrt{n} \varphi_{n-1}$, $\hat{a}^\dagger \varphi_n = \sqrt{n+1} \varphi_{n+1}$ (Eq. (9.4.26)), $[\hat{a}, \hat{a}^\dagger] = 1$ (Eq. (9.3.10)).

(1) Normalization

Expanding, $e^{\lambda \hat{a}^\dagger} \varphi_0 = \sum_{n \geq 0} \frac{\lambda^n}{n!} (\hat{a}^\dagger)^n \varphi_0 = \sum_{n \geq 0} \frac{\lambda^n}{\sqrt{n!}} \varphi_n$. Hence

$$\langle \psi_\lambda, \psi_\lambda \rangle = |N|^2 \sum_{n \geq 0} \frac{|\lambda|^{2n}}{n!} = |N|^2 e^{|\lambda|^2} \stackrel{!}{=} 1 \implies \boxed{N = e^{-|\lambda|^2/2}}. \quad (1)$$

(2) Action of $\hat{a}$

Since $[\hat{a}, e^{\lambda \hat{a}^\dagger}] = \lambda e^{\lambda \hat{a}^\dagger}$ and $\hat{a}\varphi_0 = 0$,

$$\hat{a}\psi_\lambda = N(e^{\lambda \hat{a}^\dagger} \hat{a} + \lambda e^{\lambda \hat{a}^\dagger}) \varphi_0 = \lambda N e^{\lambda \hat{a}^\dagger} \varphi_0 \implies \boxed{\hat{a}\psi_\lambda = \lambda \psi_\lambda}. \quad (2)$$

The coherent state is an eigenstate of the annihilation operator with eigenvalue λ (and $\langle \psi_\lambda | \hat{a}^\dagger = \lambda^* \langle \psi_\lambda |$).

(3) Energy expectation value

With $\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2})$,

$$\langle \hat{a}^\dagger\hat{a} \rangle = \langle \hat{a}\psi_\lambda, \hat{a}\psi_\lambda \rangle = |\lambda|^2, \quad \boxed{\langle \hat{H} \rangle_{\psi_\lambda} = \hbar\omega(|\lambda|^2 + \frac{1}{2})}. \quad (3)$$

(4) Energy uncertainty

$\hat{H}^2 = (\hbar\omega)^2(\hat{N}^2 + \hat{N} + \frac{1}{4})$ with $\hat{N} = \hat{a}^\dagger\hat{a}$. Using $\hat{N}^2 = \hat{a}^\dagger\hat{a}^\dagger\hat{a}\hat{a} + \hat{a}^\dagger\hat{a}$ and $\hat{a}^2\psi_\lambda = \lambda^2\psi_\lambda$,

$$\langle \hat{N}^2 \rangle = \langle \hat{a}^2\psi_\lambda, \hat{a}^2\psi_\lambda \rangle + |\lambda|^2 = |\lambda|^4 + |\lambda|^2. \quad (4)$$

Therefore

$$\langle \hat{H}^2 \rangle = (\hbar\omega)^2(|\lambda|^4 + 2|\lambda|^2 + \frac{1}{4}), \quad \langle \hat{H} \rangle^2 = (\hbar\omega)^2(|\lambda|^4 + |\lambda|^2 + \frac{1}{4}), \quad (5)$$

$$(\Delta E_\lambda)^2 = \langle \hat{H}^2 \rangle - \langle \hat{H} \rangle^2 = (\hbar\omega)^2|\lambda|^2 \implies \boxed{\Delta E_\lambda = \hbar\omega|\lambda|}. \quad (6)$$

(5) Differential equation and its solution

In position space, from Eq. (9.3.11) with $u = x/L_0$, $\hat{a} = \frac{1}{\sqrt{2}}(u + \frac{d}{du})$. The eigenvalue equation $\hat{a}\psi_\lambda = \lambda\psi_\lambda$ gives

$$\boxed{\frac{d\psi_\lambda(u)}{du} = (\sqrt{2}\lambda - u)\psi_\lambda(u)}. \quad (7)$$

Integrating,

$$\psi_\lambda(u) \propto \exp\left(\sqrt{2}\lambda u - \frac{u^2}{2}\right). \quad (8)$$

(6) Displaced ground state

The ground state is $\varphi_0(u) \propto e^{-u^2/2}$ (Eq. (9.2.40)). Completing the square,

$$\sqrt{2}\lambda u - \frac{u^2}{2} = -\frac{1}{2}(u - \sqrt{2}\lambda)^2 + \lambda^2, \quad (9)$$

so up to a constant

$$\psi_\lambda(u) \sim e^{-\frac{1}{2}(u - \sqrt{2}\lambda)^2} = \varphi_0(u - b), \quad \boxed{b = \sqrt{2}\lambda}. \quad (10)$$

(7) Interpretation

For real λ , the state is the ground state rigidly displaced in position by b in the variable u , i.e. by

$$\Delta x = L_0 b = \sqrt{2}\lambda L_0 = \lambda \sqrt{\frac{2\hbar}{m\omega}} \quad (\langle \hat{x} \rangle = \sqrt{2}L_0 \text{Re } \lambda). \quad (11)$$

For complex λ , $\text{Re } \lambda$ sets this position displacement, while $\text{Im } \lambda$ gives the packet a nonzero *average momentum*,

$$\langle \hat{p} \rangle = \sqrt{2} \frac{\hbar}{L_0} \text{Im } \lambda, \quad (12)$$

i.e. the imaginary part launches the (otherwise undistorted, minimum-uncertainty) Gaussian into motion.

Checks

- $\lambda \rightarrow 0$: $N \rightarrow 1$, $\psi_\lambda \rightarrow \varphi_0$, $\langle \hat{H} \rangle \rightarrow \frac{1}{2}\hbar\omega$, $\Delta E \rightarrow 0$ — recovers the ground state.
- Minimum uncertainty: $\hat{a}\psi_\lambda = \lambda\psi_\lambda$ implies $\Delta x \Delta p = \hbar/2$ for all λ (coherent states saturate the bound).
- $\Delta E = \hbar\omega|\lambda|$ grows linearly in $|\lambda|$ while $\langle \hat{H} \rangle \sim \hbar\omega|\lambda|^2$, so the *relative* spread $\Delta E/\langle \hat{H} \rangle \rightarrow 0$ for large $|\lambda|$ (classical limit).